

Math 520
Homework 5

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Section 26 (Pages 170 - 172): 1, 2a, 5, 6, 7
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Section 26

Exercise 1.

- (a) Let T and T' be two topologies on the set X ; suppose that $T' \supset T$. What does compactness of X under one of these topologies imply about compactness under the other?
- (b) Show that if X is compact Hausdorff under both T and T' , then either T and T' are equal or they are not comparable.

Exercise 2(a). Show that in the finite complement topology on $\mathbb{R}$, every subspace is compact.

Proof. Let $A \subseteq \mathbb{R}$. Suppose $\{U_i\}_{i \in I}$ is an open cover of A . Note that the complement of each open cover will be a finite set. Consider the set $A \setminus U_j$. If $A \setminus U_j = \emptyset$, then U_j completely contains A , hence it is a finite subcover of $\{U_i\}_{i \in I}$. If $A \setminus U_j \neq \emptyset$, then $A \setminus U_j$ must be finite, as $A \setminus U_j = A \cap U_j^c$. Let $\{x_1, \dots, x_n\} := A \setminus U_j$. Since each $x_1, \dots, x_n \in A \subseteq \bigcup_{i \in I} U_i$, we can find $U_1, \dots, U_n \in \{U_i\}_{i \in I}$ such that $x_j \in U_j$ for each $j = 1, \dots, n$. Then by construction $A \cap \bigcap_{j=1}^n U_j^c = \emptyset$, hence $A \subseteq \bigcup_{j=1}^n U_j$. Thus A is compact. $\square$

Exercise 5. Let A and B be disjoint compact subspaces of the Hausdorff space X . Show that there exist disjoint open sets U and V containing A and B , respectively.

Proof. For each $b \in B$, there exists opens sets U_b, V_b such that $b \in U_b$, $A \subseteq V_b$, and $U_b \cap V_b = \emptyset$. Consider the cover $B \subseteq \bigcup_{b \in B} U_b$. Since B is compact, there exists a finite subcover such that $B \subseteq \bigcup_{i=1}^n U_{b_i}$. Since $A \subseteq V_{b_i}$ for each b_i , then $A \subseteq \bigcap_{i=1}^n V_{b_i}$. All together, we've found open sets $\bigcup_{i=1}^n U_{b_i}, \bigcap_{i=1}^n V_{b_i}$ such that $A \subseteq \bigcap_{i=1}^n V_{b_i}$, $B \subseteq \bigcup_{i=1}^n U_{b_i}$, and $(\bigcap_{i=1}^n V_{b_i}) \cap (\bigcup_{i=1}^n U_{b_i}) = \emptyset$. $\square$

Exercise 6. Show that if $f : X \rightarrow Y$ is continuous, where X is compact and Y is Hausdorff, then f is a closed map.

Exercise 7. Show that if Y is compact, then the projection $\pi_1 : X \times Y \rightarrow X$ is a closed map.

Section 26

Exercise 2.

(a)

(b)

Exercise 3.

Exercise 4.